

**Ridge, LASSO, and Elastic Net Regression: Fathers' Psychological Well-being in Familial
Contexts**

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Abstract

This study examined the applicability of regularized regression in family science by focusing on fathers' psychological well-being as an applied example. Using cross-sectional survey data from 277 U.S. fathers, we compared Ordinary Least Squares (OLS) regression with Ridge, LASSO, and Elastic Net in regularized regression to evaluate how diverse familial factors—including father involvement, father role identity, parenting competency, parent–child relationships, coparenting relationships, and work–family conflict—concurrently predict fathers' well-being. Results indicated that while the three regularized approaches produced similar overall performance, they provided advantages over OLS by reducing overfitting, improving generalizability, and enhancing interpretability. We also compared the analysis results of the three approaches in terms of the prediction accuracy and interpretability. The findings reveal how regularized regression can highlight key factors, such as father involvement, coparenting, and parenting competency. This study also illustrates how regularized regression complements traditional regression in family science by addressing multicollinearity, reducing overfitting, and supporting feature selection. These results highlight the potential of computational approaches such as machine learning to strengthen both theory and practice in family research.

Keywords: Father involvement, fathers' psychological well-being, regularized regression, supervised machine learning

Ridge, LASSO, and Elastic Net Regression: Fathers' Psychological Well-being in Familial Contexts

Family science research often includes multiple, interrelated research factors such as well-being, relationship quality, and parenting practices (e.g., Baldwin et al., 2018; Nomaguchi & Milkie, 2020; Sarkadi et al., 2008). However, traditional regression methods can run into problems such as multicollinearity, overfitting, and limited interpretability, especially when predictors are correlated or when the goal is to identify the most influential factors from a larger set of variables (James et al., 2022). These methodological challenges may have contributed to the replication crisis, which highlights that many psychological studies are difficult to replicate or yield results that differ from the original findings (McNeish, 2015; Yarkoni & Westfall, 2017). For instance, among 100 experimental and correlational studies published in psychology journals, only 36% were successfully replicated (Open Science Collaboration, 2015).

Overfitting is one of the reasons for this replication issue; it occurs when a model fits the data at hand exceptionally well but performs poorly on unseen or new data because it lacks generalization (McNeish, 2015). Overfitting can be understood in terms of the concepts of bias and variance in supervised machine learning (Géron, 2019). Bias refers to the systematic error introduced by assuming a simplified model representing the data's underlying patterns. Variance reflects how much the performance of a model can vary when different datasets are used.

Overfitting can occur when the fitted model performs very well on the sample (low bias) but poorly on new, unseen data (high variance), which can be a barrier to generalizing the research findings. Ordinary Least Squares (OLS) regression, one of the most widely used statistical analysis methods, may encounter overfitting issues because it minimizes the error between predicted and observed values in the training data, potentially capturing noise along with the true

signal (McNeish, 2015; Rajula et al., 2020). Overfitting is particularly problematic when the number of predictors approaches or exceeds the number of observations, leading to inflated regression coefficients, underestimated standard errors, and overly optimistic R^2 (coefficient of determination) values that fail to generalize well to new data (McNeish, 2015). These limitations highlight the need for analytic approaches that prioritize generalizability and interpretability, which methods like regularized regression in machine learning can offer. They can also serve as tools for extending and refining theory in the social sciences, creating new opportunities for advancing family science research (Sun, 2024).

The primary goals of machine learning are the generalization of new data and prediction accuracy, which contrasts with traditional statistical approaches that focus on hypothesis testing and parameter estimation (Rajula et al., 2020). Supervised machine learning involves training algorithms on labeled data, where each input is paired with a known output (Sun, 2024). For example, in a dataset predicting fathers' psychological well-being, labeled data may include inputs like "father involvement scores" alongside the known outcome of "psychological well-being scores." The model learns from these labeled examples to recognize patterns and then makes predictions on new, unseen data (Jiang et al., 2020). Regularized regression, a component of supervised machine learning, incorporates a penalty term into a regression's objective function to prevent overfitting and enhance the regression model's performance and generalization (Jacobucci et al., 2023). Regularized regression also helps manage multicollinearity, supports variable selection to produce more parsimonious models, and improves interpretability in research settings (Jacobucci et al., 2023; McNeish, 2015). These benefits can be especially relevant to family science, where a broad set of familial and sociodemographic variables are often intercorrelated (Chan et al., 2022; Hamon, 2024).

As an applied example of regularized regression, we focused on fathers' psychological well-being—an important yet understudied topic in family research—using a comprehensive set of predictors, given that fathers' psychological well-being plays a crucial role in family and parenting quality within family dynamics. Psychological well-being refers to an individual's overall mental and emotional health, encompassing aspects such as life satisfaction, positive relationships, life purpose, and personal growth (Ryff, 2014). Fathers experiencing challenges in psychological well-being often show mood instability, irritability, and reduced emotional connection, which can negatively affect interactions with their children (Fisher, 2017). For instance, Hickey et al. (2020) found that fathers' poor psychological well-being, including parental stress, was related to an increased risk of criticism toward their children. Regarding the importance of fathers' psychological well-being, previous studies have highlighted familial factors such as father involvement, parental competency and self-efficacy, relationship quality with children, coparenting dynamics within families, and family cohesion among different samples with diverse racial/ethnic, socioeconomic, and cultural backgrounds (Bamishigbin et al., 2020; Boyraz & Sayger, 2011; Hickey et al., 2020; McGill, 2014; Varga & Gee, 2017; Tan et al., 2024). However, there have been mixed findings regarding the identification of its more influential predictors among familial factors. To illustrate, some studies emphasize father involvement as an influential factor (Garfield et al., 2010; McGill, 2014), while others highlight coparenting quality (Price-Robertson et al., 2017; Schulz et al., 2023), parent-child relationship quality (Nomaguchi & Milkie, 2020; Tan et al., 2024), or work-family conflict (Cooklin, 2019). These mixed findings may raise concerns about their generalizability, which may also be related to the replication crisis in social science.

Responding to these challenges, this study applied regularized regression techniques to demonstrate their use in family science, using fathers' psychological well-being as an example. We examined how diverse familial factors—father involvement, father role identity, parental competency, parent–child relationships, coparenting relationships, and work–family conflict—concurrently predict fathers' psychological well-being with cross-sectional data. We also compared Ridge, LASSO (Least Absolute Shrinkage and Selection Operator), and Elastic Net analysis results to illustrate how these techniques can be interpreted and applied in similar research. Given their purpose of preventing overfitting and improving generalization, regularized regression methods can offer more robust findings for researchers, counselors, and educators seeking to support fathers' psychological well-being.

Regularized Regression Approach

Traditional social science research typically reserves *prediction* to longitudinal designs in which predictors precede outcomes. However, machine learning typically uses the term to refer to modeling relationships between variables regardless of temporal order (Yarkoni & Westfall, 2017). In supervised machine learning, regularized regression is a method for predicting a continuous outcome variable, such as psychological well-being, from input variables (features). During training, the model learns patterns in the data by linking these features (e.g., father involvement, coparenting relationships) with known outcomes (labels), which are the observed values of the outcome variable. Regularized regression minimizes prediction errors, as captured in a loss function, while applying a penalty term to prevent overfitting (Aheto et al., 2021).

There are three categories of regularized regression: Ridge (L2 regularization), LASSO (L1 regularization), and Elastic Net (a combination of L1 and L2 regularization), based on the types of penalty. The penalty term in regularized regression can take two major forms: L1 and L2

penalties. The L1 penalty (used in LASSO regression) adds the absolute values of the regression coefficients to the loss function, which encourages some coefficients to shrink to exactly zero. This enables LASSO to select only the most relevant predictors, making it a useful tool for feature selection. On the other hand, the L2 penalty (used in Ridge regression) adds the squared values of the regression coefficients to the loss function, which shrinks all coefficients toward zero but does not eliminate any completely. This helps stabilize the model, particularly in cases of multicollinearity—a situation where predictor variables are highly correlated, making it difficult to separate their individual effects on the outcome. Elastic Net combines both L1 and L2 penalties, balancing feature selection and stabilization. Once trained, the model uses the patterns it has learned to make accurate predictions for new, unseen data (Usman et al., 2021).

Ridge Regression

Ridge regression, introduced by Hoerl and Kennard (1970), is a regularization method used in linear regression to address issues like multicollinearity and overfitting. It modifies the OLS regression by adding an L2 penalty term to the cost function, which shrinks the regression coefficients toward zero but never sets them exactly to zero, thereby retaining all predictors in the model (Ogutur et al., 2012). The objective function of Ridge regression is given by

$$\hat{\beta}_{ridge} = \arg \min \left[\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ij} \right)^2 + \left(\sum_{j=1}^p \beta_j^2 \right) \right]$$

The function *argmin* returns the inputs for which the outcome is minimal, where y_i is an observed response for observation i , X_{ij} is a predictor variable j for observation i , β_j is a regression coefficient, and $\hat{\beta}$ is a regression coefficient estimate. λ is the regularization tuning parameter that controls the strength of the penalty term. Larger λ values result in greater shrinkage of the coefficients. When λ equals zero, Ridge regression reduces to OLS regression;

however, as λ increases, the regression coefficients shrink toward zero but do not reach exactly zero, meaning all predictors are retained. The intercept term β_0 was excluded from penalization, consistent with standard implementations of regularized regression. With the L2 penalty term, Ridge regression can handle multicollinearity effectively by stabilizing coefficient estimates. It can also reduce overfitting, improving prediction accuracy on test data, and works well in high-dimensional settings where the number of predictors (p) exceeds that of observations (n).

LASSO Regression

LASSO regression, proposed by Tibshirani (1996), is a regularization method designed to enhance prediction accuracy and model interpretability by performing both coefficient shrinkage and variable selection. Unlike Ridge regression, which uses an L2 penalty, LASSO applies an L1 penalty, which can shrink some coefficients to exactly zero, effectively excluding irrelevant predictors from the model. The objective function for LASSO regression is:

$$\hat{\beta}_{LASSO} = \arg \min \left[\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right]$$

In contrast to the L2 penalty, the L1 penalty introduces a constraint that forces some coefficients to be exactly zero, resulting in a sparse model where irrelevant predictors are excluded. LASSO regression is effective in high-dimensional datasets with correlated predictors because it eliminates irrelevant features, resulting in a more straightforward and interpretable model (Usman et al., 2021). LASSO regression is a powerful tool for handling large datasets where feature selection is crucial, allowing for improved prediction accuracy and model interpretability (McNeish, 2015).

Elastic Net

Elastic Net is a regularization technique that combines the L1 (LASSO) and L2 (Ridge) penalties. It was introduced to address the limitations of both methods, particularly in datasets with highly correlated predictors, where LASSO alone may arbitrarily select one predictor and exclude others (Ogutur et al., 2012; Usman et al., 2021). The objective function for Elastic Net is:

$$\hat{\beta}_{Elastic} = \arg \min \left[\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ij} \right)^2 + \lambda \left[(1 - \alpha) \sum_{j=1}^p \beta_j^2 + \alpha \sum_{j=1}^p |\beta_j| \right] \right]$$

In Elastic Net, the tuning parameter λ controls the overall strength of regularization. A mixing parameter α is often introduced to balance the L1 and L2 penalties: when $\alpha=1$, Elastic Net reduces to LASSO, and when $\alpha=0$, Elastic Net reduces to Ridge. Elastic Net regression offers a robust framework that leverages the strengths of both LASSO and Ridge, making it an ideal choice for complex, high-dimensional datasets where feature selection and regularization are crucial (Jacobucci et al., 2023).

In practice, regularized regression models are trained through an iterative process that adjusts key parameters (e.g., tuning and mixing values) to minimize prediction error. This is typically performed using k-fold cross-validation, in which the data are repeatedly split into training and validation sets to evaluate performance across partitions. These steps help improve generalization, reduce overfitting, and produce more robust predictions (Géron, 2019). Such methodological strengths also align with broader calls in computational social science for methods that yield findings that are more robust, generalizable, and interpretable, thereby contributing to theory development in fields such as family science (Sun, 2024).

Fathers' Psychological Well-being in the Familial Context

Theoretical Framework

Grounded in family systems theory (Cox & Paley, 1997), fathers' psychological well-being is closely tied to family dynamics and external stressors, with these elements influencing one another reciprocally within the family context. First, fathers' psychological well-being can also play a crucial role in fostering a healthy home environment and promoting good communication, mutual respect, and emotional support among family members (Boyras & Sayger, 2011; Fisher, 2017; Hickey et al., 2020). Fathers with positive psychological well-being are more likely to provide their children with emotional stability, guidance, and encouragement. Fathers also tend to communicate more effectively with their partners and provide greater emotional support to them (Boyras & Sayger, 2011). Fathers with lower levels of psychological well-being may experience mood fluctuations, irritability, and reduced emotional connection, which negatively affect parent-child relationships (Fisher, 2017).

According to family systems theory (Cox & Paley, 1997), families function as interconnected systems in which each member's behavior and well-being affect the others. Consequently, fathers' psychological well-being is shaped by interactions within the family unit and their roles in family functioning. Moreover, challenges outside the family, such as work-family conflict, can impact the family as a whole. These conflicts may create strain within the family system, increasing stress and reducing fathers' availability for family interactions. Supportive of this theoretical perspective, empirical research has provided evidence that various familial factors can influence fathers' psychological well-being, including their involvement with children (Garfield et al., 2010), parenting competence (Giallo et al., 2013), the quality of their relationships with their children (Tan et al., 2024), coparenting dynamics (Price-Robertson et al., 2017), and work and life challenges (Cooklin, 2019).

Familial Factors of Fathers' Psychological Well-being

Father Involvement

Father involvement encompasses various dimensions, including direct engagement with their child, taking responsibility for child-related duties, and monitoring their child's activities and social interactions (Parke, 2000). Fathers' involvement with their children can be linked to their psychological well-being. Engaged fathers often lead healthier lives, reducing risky behaviors and increasing their use of health services (Garfield et al., 2010). Active involvement in their children's lives aligns with societal expectations of paternal responsibility, fostering admiration and respect from others, which can enhance fathers' confidence and psychological well-being (McGill, 2014). Also, meeting societal expectations of being an involved father can lead to admiration and respect from others, fostering a sense of fulfillment and self-worth in fathers, which can contribute to their psychological well-being (Genesoni & Tallandini, 2009; Ihinger-Tallman et al., 1993).

Father Role Identity

Father identity refers to how a father perceives his role in social relationships, influenced by societal expectations and the significance he places on that role in his life (Goldberg, 2013; Thoits, 2012). Identity theory, often used in fathering research, views identity as a person's self-conception of their role within social relationships (Stryker, 1968). This approach views identity as internalized role expectations, with individuals possessing multiple identities that vary in importance, referred to as role identity salience (Stryker & Burke, 2000). Role identities provide purpose and meaning, which can enhance well-being. Thoits (2012) found that role identity salience among volunteers was linked to better mental and physical health, mediated by a sense of purpose and meaning. In line with this, a systematic review of new fathers'

experiences reveals that the formation of a fatherhood identity has impacts on the mental health and well-being of first-time fathers (Baldwin et al., 2018).

Fathers' Competency as a Parent

Fathers' confidence in their parenting abilities can be closely linked to their psychological well-being. Bamishigbin et al. (2020) found that higher parental competency was significantly associated with lower depressive symptoms in fathers, likely because greater competency enhances parenting satisfaction, which in turn positively influences psychological health. Philpott et al. (2017) reported that feelings of incompetence in infant care during the perinatal period contributed to stress, as fathers may feel inadequate or unable to meet their children's needs. Similarly, Giallo et al. (2013) observed that fathers with low parental competency during the postnatal period were more prone to psychological distress, suggesting that challenges like managing an infant's sleep can undermine fathers' sense of competence and negatively impact their psychological well-being.

Parent-Child Relationships

Parenting is both challenging and rewarding, as parental responsibilities can be stressful yet also bring joy and happiness (Musick et al., 2016). The quality of parent-child relationships is closely tied to parents' psychological well-being. For instance, the demands of parenting, which can lead to conflicts in parent-child relationships, are negatively associated with parents' psychological well-being (Nomaguchi & Milkie, 2020). On the other hand, positive parent-child relationships can significantly enhance the psychological well-being of parents, as they foster happiness and fulfillment (Tan et al., 2024).

Coparenting Relationships

Coparenting refers to the ways in which parents relate to each other in their parental roles (Feinberg, 2003), and these relationships play a significant role in fathers' psychological well-being. Varga and Gee (2017) found that better coparenting quality was associated with increased paternal involvement, which can positively impact fathers' psychological well-being (McGill, 2014). Schulz et al. (2023) found that supportive coparenting was tied to fewer psychological well-being challenges in fathers, which also reduced parenting hostility. Price-Robertson et al. (2017) reported a strong association between coparenting quality and fathers' psychological well-being, even after accounting for factors like parental relationship happiness and maternal psychological well-being.

Work-Family Conflict

Work-conflict arises when the demands of work and family roles are incompatible, leading to stress and strain (Carlson et al., 2000). Irregular work schedules, low job satisfaction, and job insecurity were indirectly linked to greater parental concern for their children's psychological well-being by contributing to higher levels of depressive symptoms in parents (Bilodeau et al., 2023). Cooklin (2019) also reported that intense work-family conflict was associated with higher psychological distress among fathers. However, fathers who managed to reduce this conflict, such as by adopting flexible work schedules, experienced significant improvements in their psychological well-being.

Sociodemographic Factors

Various sociodemographic factors, including age, race/ethnicity, education, income, employment status, and relationship status can influence fathers' psychological well-being. Older fathers may be better equipped to manage parenting challenges than their younger counterparts (Castillo et al., 2011). Black and Hispanic fathers are more likely than White fathers to find

parenthood a source of happiness and life satisfaction (Milkie et al., 2008), but other studies also reported that Hispanic fathers reported increased depressive symptoms from parental stress (Garfield et al., 2018). Additionally, fathers with higher educational attainment and income levels often experience better psychological well-being, as these resources can alleviate economic strains and support active involvement in their children's lives (Castillo et al., 2011). Employment provides stability and financial security, contributing positively to fathers' psychological well-being (Schindler, 2010). Moreover, a stable relationship status enhances father involvement and benefits psychological well-being (Waldvogel & Ehlert, 2016). While having more children can be linked to more parental stress and can deteriorate fathers' psychological well-being (Hong & Liu, 2019), the children's co-residential status with their father is also worth considering, as nonresident fathers are at greater risk of depressive symptoms than resident fathers (Kotila & Kamp Dush, 2013). Child age can be another consideration, given its developmental stages, such as infants and toddlers, where parental demands are usually higher and bring higher parenting stress (Goodman et al., 2011).

Current Study

Regularized regression techniques, such as Ridge, LASSO, and Elastic Net, can provide a valuable approach for family science research when examining multiple, and often correlated, predictors of key outcomes (Rajula et al., 2020). These methods, commonly applied in data science and machine learning, address limitations of traditional regression—such as overfitting, reduced predictive accuracy, and limited generalizability—that have been connected to the replication crisis in the social sciences. Although these machine learning techniques have been increasingly recognized across disciplines, their application in family science, including research on fathers' psychological well-being, remains limited (McNeish, 2015; Sun, 2024).

The current study aims to demonstrate the application of three regularized regression techniques using fathers' psychological well-being as an applied example. Fathers' psychological well-being is a meaningful construct in family research because it relates to parenting quality, family functioning, and overall family health. Using this topic as a case example, the data for this study encompass a comprehensive set of familial factors potentially related to fathers' well-being in familial contexts, including father involvement, father role identity, parental competency, parent-child relationships, coparenting relationships, work-family conflict, and fathers' sociodemographic characteristics. This dataset is well-suited because it captures multiple aspects of family dynamics, reducing the risk of replication challenges that often stem from inconsistent sets of independent variables across studies (Yarkoni & Westfall, 2017). Second, fathers are treated as the focal individuals in this dataset. In much of developmental and family research, fathers are often considered primarily as the secondary caregiver to the focal child or the partner of the focal mother (Genesoni & Tallandini, 2009), which frequently results in substantial missing data for father variables. In contrast, the present dataset places fathers at the center of measuring family dynamics, allowing for a more comprehensive and informative analysis.

The specific goals of this study are: (1) to conduct an OLS regression as a baseline model and to apply Ridge, LASSO, and Elastic Net regression techniques to investigate fathers' psychological well-being regarding the familial factors, with attention to preventing overfitting and improving the generalization, and (2) to compare the results of these three approaches in terms of prediction accuracy and interpretability. This study illustrates how regularized regression can be applied in family science research, demonstrating its usefulness for addressing

multicollinearity, reducing overfitting, improving generalizability, and selecting variables to build more interpretable models.

Methods

Participants

Data were collected using the Qualtrics survey panel from December 2020 to June 2021. Participants were recruited through various methods, such as emails, website pop-ups, social media ads, and Qualtrics partner organizations across the U.S. Interested individuals accessed the survey link, reviewed the consent form detailing the study's purpose and eligibility criteria, and provided consent before confirming their eligibility. Eligible participants were men aged 18 or older with at least one child under the age of 18. This study was approved by the institutional review board of a Southeastern University.

The final sample consisted of 277 U.S. fathers, with a mean age of 37.62 years ($SD = 9.46$, range: 19–72). The racial composition was 59.6% White, 16.6% Hispanic/Latino, and 13.0% African American. 34.3% of the participants fell into the 'High school/GED or less education' category, and 30.7% had a graduate degree. 18.1% of the sample earned \$140,000 or more, and 17.7% earned between \$80,000 and \$99,999. Regarding relationship status, 71.8% were first married, followed by single (13.0%) and cohabiting (5.8%). Most participants (78.7%) worked full-time. The average number of children living together was 1.78 ($SD = 1.05$), and that of not living together was 0.20 ($SD = 0.69$). The average number of children aged 0 to 6 (early childhood) was 0.96 ($SD = 0.81$), and that of children aged 7 to 12 (middle childhood) and 13 to 17 (adolescence) was 0.65 ($SD = 0.74$) and 0.37 ($SD = 0.71$), respectively. Table 1 provides further sociodemographic details.

[Table 1 around here]

Measurements

Psychological Well-being

Fathers' psychological well-being was assessed using the 18-item Psychological Wellbeing (PWB) scale (Ryff & Keyes, 1995). Participants rated items on a seven-point scale (1 = strongly agree, 7 = strongly disagree), with some reverse-coded so higher scores reflected greater well-being. Example items include "*I like most parts of my personality*" and "*I am good at managing the responsibilities of daily life.*" Cronbach's alpha was .74, and the average score was used for analysis.

Father Involvement

The 35-item Father Involvement Scale (Hawkins et al., 2002) measured father involvement, with fathers rating their parenting activities over the past 12 months on a 0–6 scale (where 0 = very poor and 6 = excellent). Example items include "*Disciplining your children*" and "*Telling your children that you love them.*" Higher scores indicated greater involvement, and the average score was used for analysis (Cronbach's alpha = .98).

Father Role Identity

Father role identity was assessed using 12 items from the Father Role Identity Salience Scale (Fox & Bruce, 2001). Responses were rated on a four-point scale (1 = not at all true of you to 5 = always true of you). Example items include "*I like being known as a father*" and "*Being a father has changed me for the better.*" Some items were reverse-coded so that higher scores indicated a greater role identity. Among the 12 items, one item had a low item-total correlation and increased the scale's reliability when deleted: "*I prefer the company of adults to spending time with kids.*" The average score, except for this item, was calculated for analysis, and Cronbach's alpha was .69.

Parental Competency

Fathers' parental competency was measured with the 17-item Parent Sense of Competency Scale (PSOC; Gibaud-Wallston & Wandersman, 1978). Items were rated on a 6-point Likert scale (1 = strongly disagree, 6 = strongly agree), with some items reverse-coded, so that higher scores indicated greater competency. Example items include *“Being a parent is manageable, and any problems are easily solved.”* Among the 17 items, two items had a low item-total correlation and increased the scale's reliability when deleted, such as *“Being a parent is manageable, and any problems are easily solved.”* The average score, excluding these items, was calculated for analysis, and Cronbach's alpha was .68.

Parent-Child Relationships

The quality of parent-child relationships was assessed using the 15-item Short Form of the Child-Parent Relationship Scale (CPRS-SF; Driscoll & Pianta, 2011), with subscales for relationship closeness (Cronbach's alpha = .88) and conflict (Cronbach's alpha = .92). Example items are such as *“My child values his/her relationship with me”* and *“Dealing with my child drains my energy.”* Participants rated items on a 5-point scale (1 = definitely does not apply, 5 = definitely applies), with averages calculated for each dimension for analysis.

Coparenting Relationships

The 35-item Coparenting Relationship Scale (CRS; Feinberg et al., 2012) was used to measure coparenting relationship quality. Responses ranged from 0 (not true) to 6 (very true), with some reverse-coded, so higher scores indicated stronger relationships. Example items include *“My partner and I have the same goals for our child”* and *“My partner undermines my parenting.”* The average score was used for analysis, and Cronbach's alpha was .79.

Work-Family Conflict

Work-family conflict was measured using the 18-item Multidimensional Measure of Work and Family Conflict (MMWFC; Carlson et al., 2000), with items rated on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). Example items include “*Due to all the pressures at work, sometimes when I come home I am too stressed to do the things I enjoy.*” Higher scores indicated more conflict, and the average score was used for analysis (Cronbach’s alpha = .95).

Sociodemographic Characteristics

Seven covariates of fathers’ psychological well-being were considered: age, race/ethnicity, education, income, work status, relationship status, the number of children living together or not, and the number of children in three age categories. Participants’ age, number of children living/not living, and number of children in three age categories were continuous. We regarded education and income as continuous. Race/ethnicity and relationship status were categorical, and we used ‘White’ and ‘First married’ as the baseline for each variable. Work status was coded as 0 (full-time) and 1 (not full-time).

Analysis plan

We used IBM SPSS 30.0 for descriptive and correlation analyses, and also Google Colab for OLS regression and regularized regression analysis (<https://colab.research.google.com/>). Google Colab is a freely available, cloud-based platform that enables users to write and execute Python code in a Jupyter Notebook environment, making it ideal for machine learning, data analysis, and collaboration. We first ran descriptive and correlation analyses for study variables. We used the *statsmodels* library to run OLS regression using the entire dataset. We then utilized the Ridge, LASSO, and Elastic Net classes from the `linear_model` module in scikit-learn to implement regularized regression models. The modeling process of regularized regression includes splitting the data into a training and test set, standardizing variables, fitting the model,

testing it, tuning hyperparameters, performing cross-validation, and evaluating model metrics.

The overall analysis procedure follows standard practices for implementing regularized regression as outlined by Jacobucci et al. (2023).

Data Preparation

To evaluate how well the model generalizes to new data, the dataset is split into two subsets: one for training the model and another for testing its predictions, which is used to evaluate the model's performance on unseen data, ensuring it generalizes well. Each dataset is also divided into predictor variables (e.g., father involvement and parent-child relationships) and the target variable (fathers' psychological well-being). Predictor variables or features should be standardized in regularized regression. As L1 (LASSO) and L2 (Ridge) penalties rely on the size of the coefficients to shrink them and reduce overfitting, features with more extensive ranges can dominate the penalty term, resulting in biased coefficient estimates.

Model Initialization and Fitting

The analysis involves applying one of three regularized regression methods: Ridge, LASSO, or Elastic Net. These methods utilize penalty terms to enhance model performance by mitigating overfitting and stabilizing regression coefficients. The selected regression method is trained on the training data to identify patterns between the predictors and the target variable. This step generates regression coefficients, which help determine the predictors most strongly associated with the target variable.

Hyperparameter Tuning and Cross-Validation

Hyperparameter tuning is conducted using grid search with cross-validation to optimize model performance. This process iteratively tests different values of the tuning (λ) and mixing parameters (α) to identify the best combination of parameters that minimizes prediction error.

The *GridSearchCV* option in the Scikit-learn machine learning library automatically searches for the best model hyperparameters in regularized regression. Cross-validation ensures the model performs consistently and generalizes well to new, unseen data. In K-fold cross-validation, the dataset is split into K equal subsets (folds); for example, with $K = 5$, the data is split into five subsets. The model is trained and validated K times, using one fold for validation and the remaining K-1 folds for training in each iteration. Model performance metrics are calculated for each run, and the final performance score is obtained by averaging the results across all K iterations, which provides a robust evaluation of model performance.

Model Evaluation

After training, the model's predictions are tested against the reserved test data. Two key metrics can be used to evaluate its performance: Root Mean Squared Error (*RMSE*) and R-squared (R^2). *RMSE* measures the average magnitude of the prediction errors, representing the difference between predicted and actual values, how far the model's predictions are from the actual values on average (Géron, 2019; Jacobucci et al., 2023). While there is no absolute cut-off for a good or bad *RMSE* value, when the *RMSE* is 0.5, the model's predictions are off by an average of 0.5 units from the true values, where "unit" indicates the scale or measurement of the target variable in the dataset. Lower *RMSE* values indicate more accurate predictions. R^2 reflects the proportion of variance in the target variable that the predictors in the model explain. R^2 values range from 0 to 1, with higher values indicating a better fit (Géron, 2019; Jacobucci et al., 2023). For example, an R^2 of .6 suggests that the model's predictors can explain 60% of the variation in the target variable.

Prediction and Interpretation

Predictions are made for the test data using the trained model with the training data set. These results are then interpreted to identify key predictors that influence the outcome of interest, providing actionable insights for research and practice.

Results

Descriptive and Correlation Analyses

Table 2 provides the results of the descriptive and correlation analyses of the study variables. Overall, study variables were correlated at the $p < .05$ or $p < .01$ level except for non-significant correlations between parent-child relationship closeness and parent-child relationship conflict. Fathers' psychological well-being was positively correlated with father involvement, father role identity, parental competency, parent-child relationship closeness, and coparenting relationships. In contrast, fathers' psychological well-being was negatively correlated with parent-child relationship conflict and work-family conflict.

[Table 2 around here]

OLS and Regularized Regression Analyses

We first conducted an OLS regression with the full dataset to examine concurrent predictors of fathers' psychological well-being. All the Variance Inflation Factor (VIF) values were below 5, indicating no problematic multicollinearity (Kutner et al., 2005). The overall model was statistically significant: $F_{(25, 251)} = 18.45, p < .001$. Significant positive predictors included father involvement and parental competency, whereas work-family conflict was a significant negative predictor of fathers' psychological well-being.

For regularized regression, the data were split into a training set (80%) and a test set (20%). All predictor variables were standardized to prevent biased coefficient estimates resulting from variable scale differences. Next, we fitted the three regularized regression models to the

training data with the best parameter estimates. The fitted model was tested using the test data set with 5-fold cross-validation. The hyperparameters of α and λ were tuned by the GridSearchCV option, automatically identifying the best model parameters. $RMSE$ and R^2 were used for model performance evaluation.

Table 3 presents the model evaluation metrics ($RMSE$ and R^2) for the OLS, Ridge, LASSO, and Elastic Net regression models. The lower the $RMSE$ value, the smaller the prediction error. R^2 ranges from 0 to 1; a higher value is preferable, as it explains more variance (Jacobucci et al., 2023). OLS yielded the lowest $RMSE$ and the highest R^2 on the training data, which may indicate overfitting. While OLS can perform well on in-sample data, it often fails to generalize to out-of-sample data, whereas regularized models often perform better. Three regularized regression models demonstrated similar fit, with LASSO regression yielding higher accuracy and greater explanatory power for predicting fathers' well-being from familial factors and sociodemographic characteristics.

[Table 3 around here]

Table 4 shows the regularized regression coefficients of Ridge, LASSO, and Elastic Net. P-values are not provided in regularized regression because the regularization process alters coefficient estimation, thereby breaking the assumptions required for calculating p-values in standard OLS regression, such as estimating precise standard errors (Jacobucci et al., 2023). Overall, the three models exhibit similar patterns in the sizes and directions of their familial factor coefficient values. Regarding absolute values, father involvement was the most influential predictor of fathers' psychological well-being, followed by parent-child relationship conflict, coparenting relationships, parental competency, work-family conflict, father role identity, and

parent-child relationship closeness. Also, parent-child relationship conflict and work-family conflict were negatively associated with fathers' psychological well-being.

Meanwhile, the coefficient value of parent-child relationship closeness was zero in LASSO and Elastic Net. This reflects the feature selection of the L1 penalty: eliminating variables with low predictive significance to improve the model's clarity and performance (Aheto et al., 2021). However, the Ridge model did not have any factors with zero coefficients, even when including sociodemographic information, which reflects the L2 penalty, only shrinking each variable's coefficient but not setting it to zero (Usman et al., 2021).

Across the sociodemographic variables, the three regularized regression approaches yielded similar patterns for participants' age, household income, and Hispanic/Latino ethnicity (compared with White). Additionally, the outcomes of the three approaches were largely consistent in terms of the number of resident and nonresident children as well as the number of children aged 7 to 12, although their regularized regression coefficients were minimal. The results for other sociodemographic variables varied across the three regularized regression analyses. In the case of LASSO regression, in particular, most variables were reduced to having no influence, as their coefficients shrank to zero.

[Table 4 around here]

Discussion

The present study explored how supervised machine learning can be applied in family science, with a focus on regularized regression as an alternative approach to addressing issues such as overfitting, which often contributes to the broader replication crisis. We introduced the concepts of three regularized regression approaches—Ridge, LASSO, and Elastic Net—and applied these techniques to examine how fathers' familial factors relate to their psychological

well-being. Regularized regression approaches can help family science researchers to address multicollinearity, reduce overfitting, and thereby improve generalizability. As an exemplary study, we conducted Ridge, LASSO, and Elastic Net regression analysis to examine how various familial factors are associated with fathers' psychological well-being. The current study's findings can provide valuable guidance to researchers and practitioners on key aspects of family life related to fathers' psychological well-being such as father involvement and coparenting relationships. These findings can also be referenced for designing intervention programs that enhance fathers' psychological well-being and, in turn, improve overall family well-being.

Applying Regularized Regression to Family Science Research

Regularized regression offers several benefits relevant to family science. By penalizing model complexity, this approach reduces overfitting and thereby improves predictions on unseen data and the generalizability of research findings, which is a core concern in the context of replication (Yarkoni & Westfall, 2017). Regularized regression also helps stabilize estimates when predictors are highly correlated, a situation often seen in family research where familial and sociodemographic factors are closely interrelated (Jacobucci et al., 2023; Hamon, 2024). Methods that impose an L1 penalty, such as LASSO or the LASSO component of Elastic Net, support variable selection and yield more parsimonious, interpretable models (Tibshirani, 1996). These benefits also support broader arguments that computational methods, including machine learning, can enhance theory in the social and health sciences by expanding the scope of data analysis and yielding findings that are more robust and generalizable (Sun, 2024).

Our inclusion of an OLS baseline can highlight these advantages. While OLS provided a helpful point of reference, it was more susceptible to overfitting when the number of predictors was numerous and they were correlated. On the contrary, the regularized models achieved

comparable or better cross-validated error while improving model parsimony (e.g., LASSO shrinking some coefficients to zero) and interpretability (McNeish, 2015). This pattern shows why and how regularization can serve as a helpful complement to OLS in family science. At the same time, the choice among Ridge, LASSO, or Elastic Net should depend on the data structure and the researcher's goals (Jacobucci et al., 2023; James et al., 2022). Ridge regression is most useful when multicollinearity is high, and researchers want to retain all predictors in the model. By shrinking coefficients toward zero, Ridge stabilizes estimates and provides more reliable interpretations in this setting (Ogutu et al., 2012). LASSO regression is preferable when feature selection and parsimony are priorities. Its L1 penalty forces some coefficients to zero, thereby reducing the model to the most influential factors. This is valuable in family science as family researchers often measure many familial and sociodemographic factors but need to isolate those that matter most (McNeish, 2015). Elastic Net, balancing Ridge's ability to handle correlated predictors with LASSO's feature selection, is beneficial when predictors are both numerous and highly correlated, making it difficult to rely on Ridge or LASSO alone (Géron, 2019).

In this study, the three approaches showed broadly similar model fits, but their differences in prediction accuracy and interpretability still offer useful insights for family science researchers. Ridge ensures stability when multicollinearity is a concern; LASSO highlights the most influential variables when model parsimony is a priority; and Elastic Net provides a compromise when both correlation and variable selection are central issues. Presenting all three models together illustrates not only their methodological utility but also how each can inform different types of research questions in family science. Some family scholars may prefer Ridge for examining theoretical models with many interrelated constructs. Others may turn to LASSO to maximize interpretability; for example, in this study, variables such as parent-child closeness

and certain sociodemographic factors were reduced to zero, directing attention to the most influential factors (Géron, 2019).

Fathers' Psychological Well-being and Familial Factors

While regularized regression has broad applications in family science, the present study's value also lies in demonstrating how these methods can be applied to a modest dataset focused on fathers' psychological well-being. For fatherhood research, our findings can illustrate how regularized regression can help identify which familial factors are most predictive of fathers' psychological well-being. Our analysis results highlight consistent trends across three models, identifying key factors to well-being while utilizing regularization techniques of Ridge, LASSO, and Elastic Net. Notably, Father involvement emerged as a robust positive predictor of psychological well-being in all three models. This finding aligns with the results of a systematic review, which identified father involvement as a significant predictor of psychological well-being (Baldwin et al., 2018). Fathers' involvement with their children is linked to better psychological well-being, promoting healthier lifestyles and increased confidence by meeting societal expectations (Garfield et al., 2010; McGill, 2014). Parent-child relationship conflict was the second most influential factor in fathers' psychological well-being: the higher the conflict level, the lower the psychological well-being. Previous research indicates that strained relationships between parents and children contribute to increased paternal stress, emotional distress, and a decline in psychological well-being (Nomaguchi & Milkie, 2020).

The next powerful predictor of fathers' psychological well-being is coparenting quality. The quality of the coparenting relationship can be another essential predictor of fathers' psychological well-being, as shown in prior work, where coparenting was positively associated with psychological well-being (Schulz et al., 2023), even when accounting for factors such as

father involvement and maternal mental health (Price-Robertson et al., 2017). Additionally, with respect to parental confidence, fathers who view themselves as competent tend to have greater confidence in managing parenting responsibilities, which confers psychological benefits (Bamishigbin et al., 2020). Another key predictor is work-family conflict, as excessive work demands lead to stress and burnout, diminishing fathers' capacity to interact with their children and contributing to tension and difficulties within the family (Carlson et al., 2000). However, the fathers' role identity was a very weak predictor of fathers' psychological well-being. This may imply that rather than identity itself having a strong influence, various other familial factors, such as coparenting quality and parental competency, play a more significant role in explaining fathers' psychological well-being.

However, parent-child relationship closeness had a minimal effect on fathers' psychological well-being, particularly compared with parent-child relationship conflict, as observed in our study. While close relationships can bring happiness (Tan et al., 2024), unresolved conflicts can cause emotional strain and stress, which in turn may negate the benefits from close relationships (Katz & Gottman, 1997). This suggests that reducing conflict may be more important than merely fostering closeness for fathers' psychological well-being.

Implications

This study highlights both substantive and methodological implications for family science. By comparing OLS with Ridge, LASSO, and Elastic Net regression, the results show that regularized regression can reduce overfitting, improve prediction accuracy, and yield more interpretable models. These advantages are particularly useful in family science, where researchers often work with interrelated predictors (Hamon, 2024). Additionally, a comparison of three approaches can serve as a reference for family science researchers who are beginning to

adopt computational methods, provided that regularization prioritizes generalization and interpretability over in-sample fit, as evaluated only on the training data. LASSO and Elastic Net, in particular, support model parsimony by identifying the most influential factors.

For practitioners and family educators, the results emphasize the importance of enhancing father involvement, strengthening coparenting relationships, and supporting parental competency to improve fathers' psychological well-being. Interventions that reduce parent–child relationship conflict and mitigate work–family conflict may also be particularly effective in fostering fathers' psychological well-being. By demonstrating the utility of regularized regression, this study also provides a methodological template for researchers and practitioners interested in applying advanced analytic approaches to identify key leverage points for supporting fathers' well-being and, by extension, overall family functioning.

Limitations

While this study offers some valuable insights, it has several limitations. First, its cross-sectional design prevents causal inference about the relationships between familial factors and fathers' psychological well-being. Future research could employ longitudinal designs and causal inference methods, such as fixed-effects and random-effects models, to provide stronger evidence regarding causal pathways (Firebaugh et al., 2013). Second, self-reported data may introduce social desirability bias, potentially overestimating the observed associations. Future studies could benefit from incorporating observational or multi-informant measures to provide a more nuanced understanding of the phenomenon. Third, the modest sample size and participants' relatively high education and income levels may limit the generalizability of the findings. Future studies with larger and more socioeconomically diverse samples would strengthen the robustness of these results. Fourth, the moderate R^2 values indicate that the current study may not have

captured all important variables or the full complexity of familial factors influencing fathers' well-being. Future research may take a more comprehensive approach and consider additional predictors, such as fathers' psychological well-being history, marital history, couple relationships, family environment, children's special needs, occupational and employment issues, and life transitions (Affleck et al., 2018; Boyraz, G., & Sayger, 2011; Norlin & Broberg, 2013), as well as interactions among familial factors, to provide a more comprehensive picture. Lastly, although we improved internal consistency by removing problematic items, the reliabilities for Family Role Identity and Parental Competency remained modest. As these lower reliabilities may have weakened the predictive power of these measures, care should be taken when interpreting the results.

Conclusion

This study examined Ridge, LASSO, and Elastic Net regression in the concurrent prediction of fathers' psychological well-being, using OLS as a baseline, and demonstrated how regularized regression can reduce overfitting, improve generalizability, and enhance interpretability in family science research. All three approaches performed similarly, with LASSO providing greater parsimony by shrinking some predictors to zero, highlighting the importance of father involvement, coparenting, and parenting competency for fathers' psychological well-being. The main contribution of this study lies in demonstrating how regularized regression can be applied to complex, interrelated family data to yield more stable and generalizable insights. Looking ahead, these methods can help family researchers manage challenges such as multicollinearity and replication, while supporting model development that informs both theory and practice.

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Table 1*Sociodemographic Characteristics (N = 277)*

Variables	<i>n</i>	%	<i>M(SD)</i>
Age	277		37.62 (9.46)
Race/ethnicity			
White	165	59.6	
African American	36	13.0	
Asian	24	8.7	
Hispanic/Latino	46	16.6	
Native Hawaiian/Pacific Islander	1	0.4	
Two or more races	5	1.8	
Education level			
High school/GED or less education	95	34.3	
Two-year college	40	14.4	
Four-year college	57	20.6	
Graduate school	85	30.7	
Household income			
Under \$ 20,000	22	7.9	
\$20,000 to \$39,999	40	14.4	
\$40,000 to \$59,999	27	9.7	
\$60,000 to \$79,999	36	13.0	
\$80,000 to \$99,999	49	17.7	
\$100,000 to \$119,999	24	8.7	
\$120,000 to \$139,999	29	10.5	
\$140,000 and more	50	18.1	
Relationship status			
First married	199	71.8	
Single	36	13.0	
Divorced	13	4.7	

Variables	<i>n</i>	%	<i>M</i> (<i>SD</i>)
Widowed	3	1.1	
Remarried	10	3.6	
Cohabited	16	5.8	
Work status			
Full time	218	78.7	
Not full-time	59	21.3	
Number of children with living status			
Living together			1.78 (1.05)
Not living together			0.20 (0.69)
Number of children in age categories			
0 ~ 6 years old			0.96 (0.80)
7 ~ 12 years old			0.65 (0.74)
13 ~ 17 years old			0.37 (0.71)

Table 2

Means, Standard Deviations, and Correlations of Study Variables (N = 277)

	1	2	3	4	5	6	7	8
1. Psychological well-being	1							
2. Father involvement	.51**	1						
3. Father role identity	.56**	.47**	1					
4. Parental competency	.55**	.15*	.57**	1				
5. P-C relationship closeness ^a	.44**	.81**	.41**	.13*	1			
6. P-C relationship conflict ^a	-.46**	.14*	-.42**	-.68**	.07	1		
7. Coparenting relationships	.59**	.56**	.53**	.32**	.49**	-.19**	1	
8. Work-family conflict	-.34**	.25**	-.33**	-.65**	.25**	.74**	-.13*	1

Mean	4.46	4.52	3.66	3.75	3.90	3.06	3.62	3.21
Standard Deviation	0.76	1.08	0.59	0.66	0.83	1.10	0.68	0.99
Scoring range	1~7	0~6	1~5	1~6	1~5	1~5	0~6	1~5

* $p < .05$, ** $p < .01$.

Note. a. P-C = Parent-child

Table 3

Regularized Regression Model Metrics of Ridge, LASSO, and Elastic Net (N = 277)

	OLS	Ridge	LASSO	Elastic Net
RMSE	0.46	0.62	0.59	0.60
R^2	.63	.31	.37	.36

Table 4

Regularized Regression Coefficients of Ridge, LASSO, and Elastic Net (N = 277)

Variables	Ridge	LASSO	Elastic Net
Father involvement	0.330	0.315	0.317
Father role identity	0.035	0.024	0.029
Parental competency	0.189	0.143	0.157
Parent-child relationship_closeness	-0.012	0	0
Parent-child relationship_conflict	-0.218	-0.219	-0.216
Coparenting relationships	0.154	0.170	0.167
Work-family conflict	-0.104	-0.101	-0.105
Age	-0.019	-0.006	-0.014
Race/ethnicity ^a			
African American	0.020	0	0
Asian	-0.087	0.000	-0.014
Hispanic/Latino	0.152	0.088	0.120

Native Hawaiian/Pacific Islander	-0.024	0	0
Two or more races	-0.191	0	0
Education level ^b	0.020	0	0.004
Household income ^b	0.041	0.019	0.025
Relationship status ^c			
Single	0.058	0	0
Divorced	-0.033	0	0
Widowed	-0.137	0	0
Remarried	0.014	0	0
Cohabited	0.202	0	0.067
Work status ^d	-0.021	0	-0.018
Number of children living together	-0.012	-0.002	-0.006
Number of children not living together	-0.005	-0.002	-0.003
Number of children aged 0 to 6	-0.012	0	-0.008
Number of children aged 7 to 12	-0.010	-0.003	-0.008
Number of children aged 13 to 17	0.002	0	0

Note. a. baseline = White; b. Education and household income level are treated as continuous; c. baseline = First married; d. Full-time (baseline) vs. Not full-time.